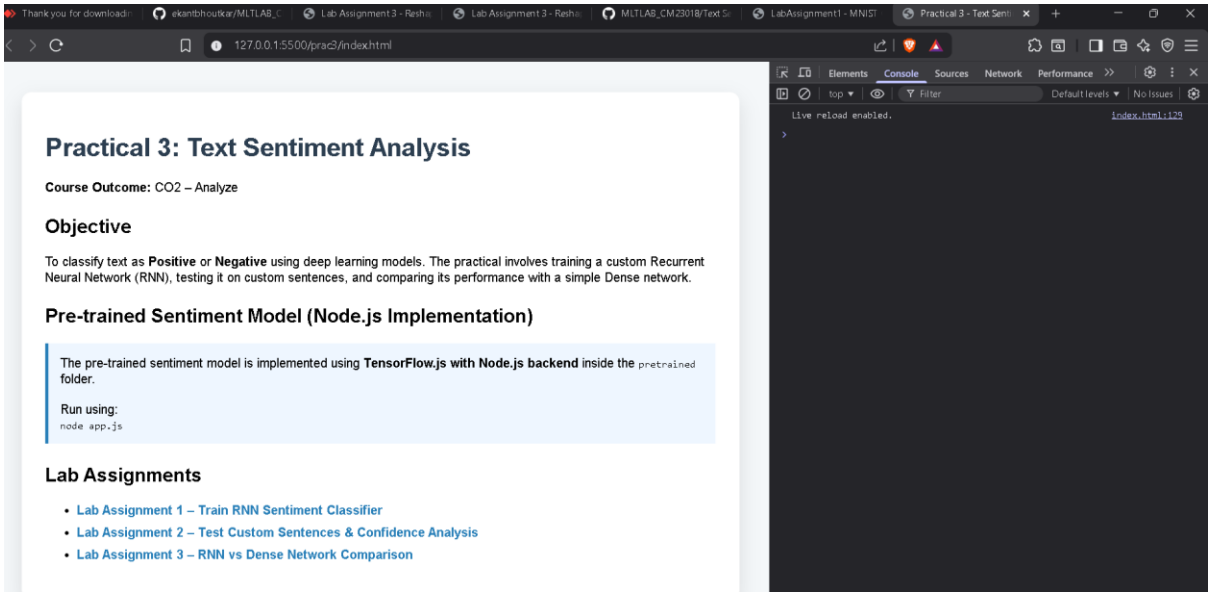
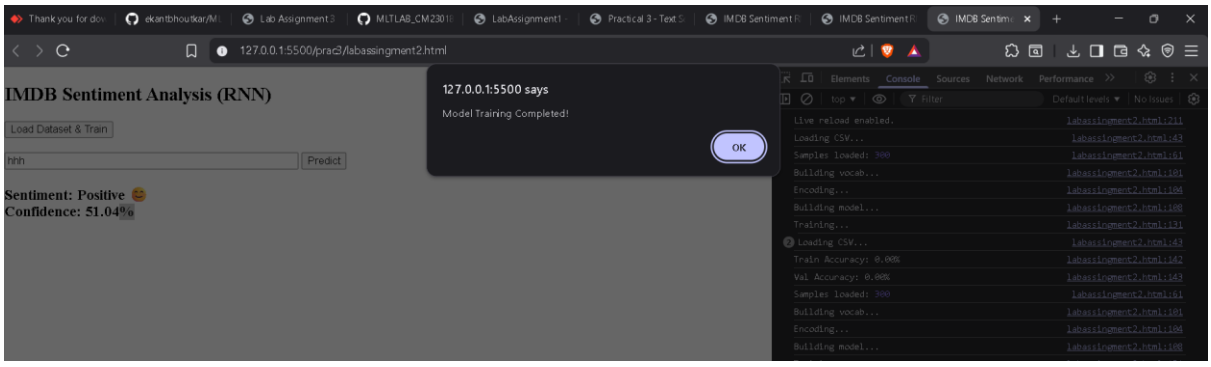
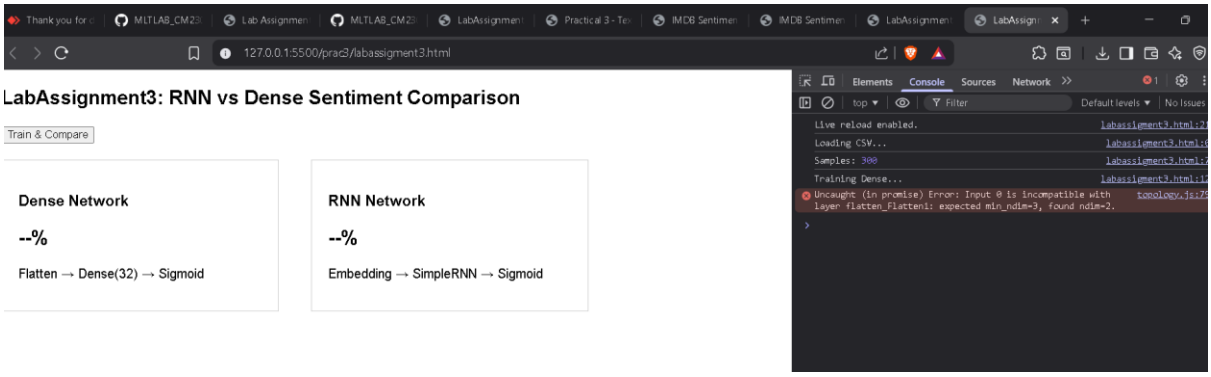

127.0.0.1:5500/prac3/index.html
Practical 3: Text Sentiment Analysis
Course Outcome: CO2 – Analyze
Objective
To classify text as Positive or Negative using deep learning models. The practical involves training a custom Recurrent Neural Network (RNN), testing it on custom sentences, and comparing its performance with a simple Dense network.
Pre-trained Sentiment Model (Node.js Implementation)
The pre-trained sentiment model is implemented using TensorFlow.js with Node.js backend inside the pretrained folder.
Run using:
node app.js
Lab Assignments
Lab Assignment 1 – Train RNN Sentiment Classifier
Lab Assignment 2 – Test Custom Sentences & Confidence Analysis
Lab Assignment 3 – RNN vs Dense Network Comparison
Elements
Console
Sources
Network
Performance
Live reload enabled.
index.html:129


127.0.0.1:5500/prac3/labassingment2.html
IMDB Sentiment Analysis (RNN)
Load Dataset & Train
hhh
Predict
Sentiment: Positive
Confidence: 51.04%
127.0.0.1:5500 says
Model Training Completed!
OK
Live reload enabled.
Loading CSV...
Samples loaded: 300
Building vocab...
Encoding...
Building model...
Training...
Train Accuracy: 0.00%
Val Accuracy: 0.00%


127.0.0.1:5500/prac3/labassigment3.html
LabAssignment3: RNN vs Dense Sentiment Comparison
Train & Compare
Dense Network
--%
Flatten → Dense(32) → Sigmoid
RNN Network
--%
Embedding → SimpleRNN → Sigmoid
Live reload enabled.
Loading CSV...
Samples: 300
Training Dense...
Uncaught (in promise) Error: Input 0 is incompatible with layer flatten_Flatten1: expected min_ndim=3, found ndim=2.